INCH-POUND

MS27245R
27 March 2012
SUPERSEDING
MS27245P
27 November 2003

DETAIL SPECIFICATION SHEET

# RELAYS, ELECTROMAGNETIC, 10 AMPERES, 2 PDT TYPE I, HERMETICALLY SEALED

INACTIVE FOR NEW DESIGN AFTER 5 JUNE 1987. NO SUPERSEDING SPECIFICATION. FOR NEW DESIGN, USE MIL-PRF-83536/9,10, 11.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-PRF-6106.

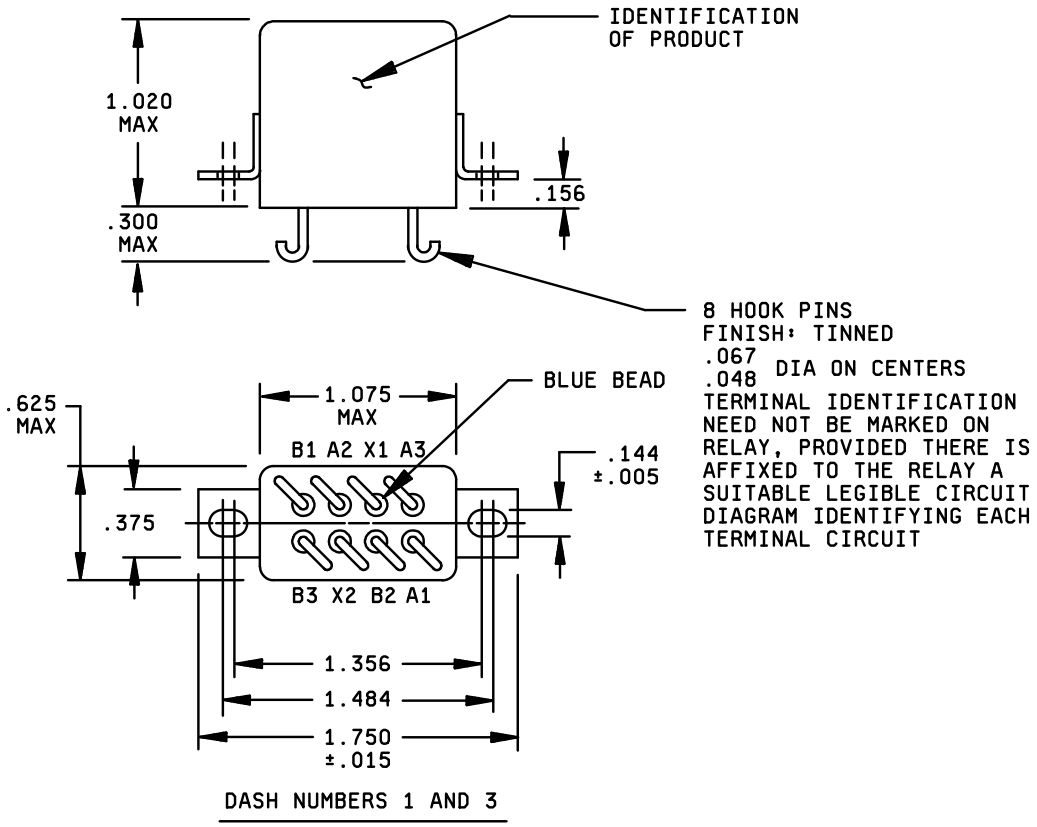



FIGURE 1. Dimensions and configuration.



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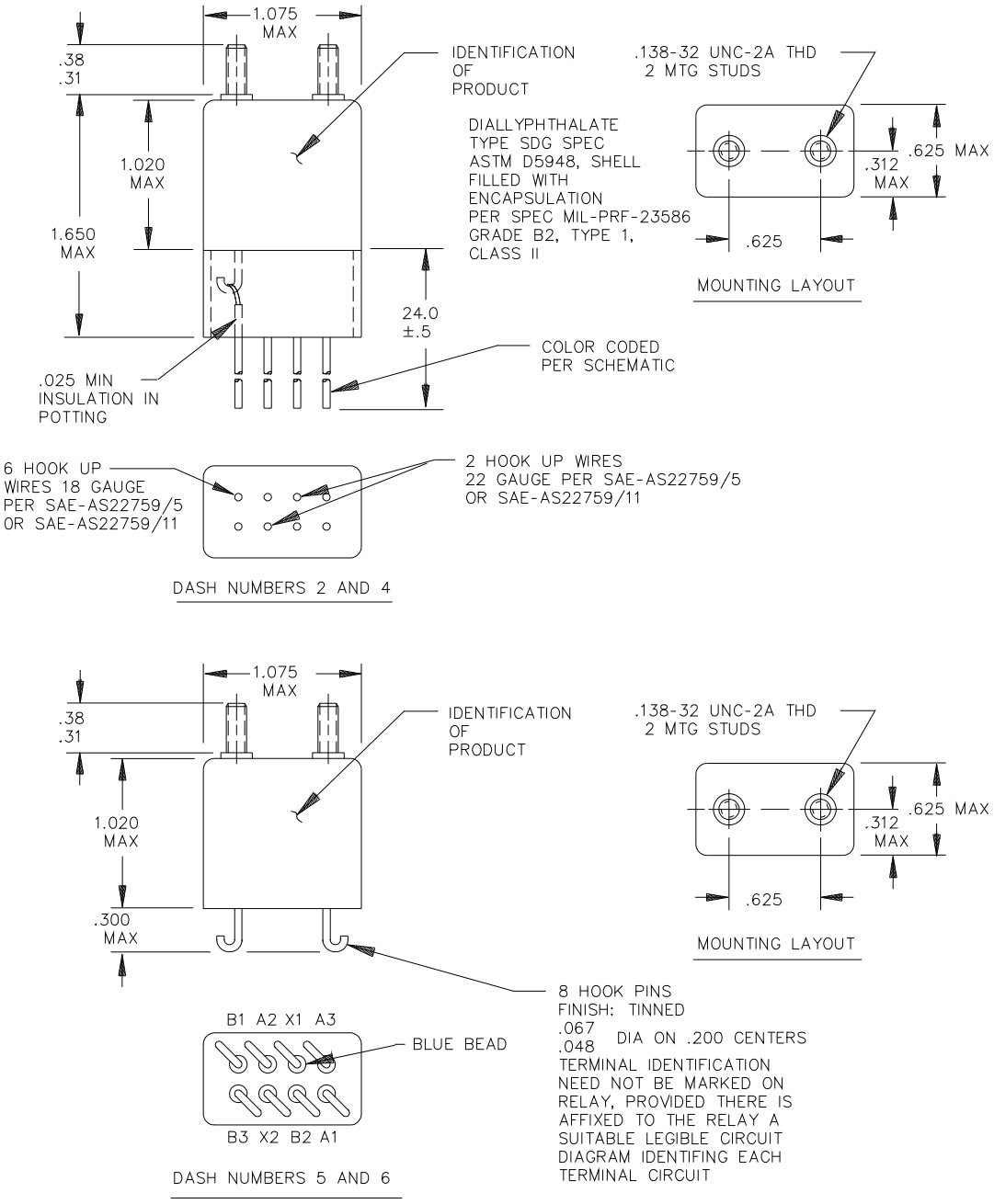
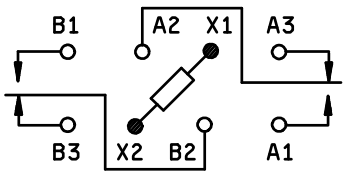


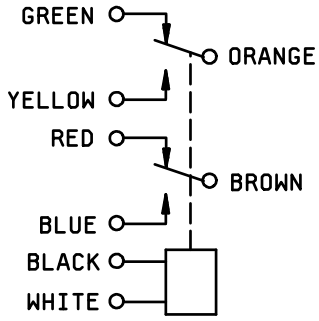

FIGURE 1. Dimensions and configurations - Continued



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CIRCUIT DIAGRAM
(DASH NUMBERS 1,3,5 AND 6)




CIRCUIT DIAGRAM
(DASH NUMBERS 2 AND 4)

| Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|
| .005 | 0.13 | .156 | 3.96 | .38 | 9.7 | 1.020 | 25.91 |
| .015 | 0.38 | .25 | 6.4 | .5 | 13. | 1.356 | 34.44 |
| .048 | 1.22 | .257 | 6.53 | .515 | 13.08 | 1.485 | 37.69 |
| .067 | 1.70 | .300 | 7.62 | .625 | 15.88 | 1.650 | 41.91 |
| .138 | 3.51 | .31 | 7.9 | 1.000 | 25.40 | 1.750 | 44.45 |
| .144 | 3.66 | .375 | 9.52 | 1.015 | 25.78 | 24.0 | 610. |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).

FIGURE 1. Dimensions and configurations - Continued.



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REQUIREMENTS:

Dimensions, and configurations: See figure 1.

ENVIRONMENTAL CHARACTERISTICS:

Temperature range: -70°C and +125°C.

Maximum altitude rating: 80,000 feet.

Shock g-level: 50 g's.

Duration: 11 ms.

Max duration contact opening: 10 μs.

Vibration - sinusoidal:

G-level: 20 g's.

Frequency range: 10 - 2,000 Hz.

Acceleration: 15 g's.

ELECTRICAL CHARACTERISTICS (SEE TABLES I, II, III, AND IV).

Insulation resistance, initial: 100 megohms.

After life or environmental tests: 50 megohms.

Dielectric strength (sea level):

| | Initial | After life tests |
|---|---|---|
| Coil to case | 1,000 V rms | 1,000 V rms |
| Aux contacts | N/A | N/A |
| All other points | 1,250 V rms | 1,000 V rms |

Dielectric strength (altitude): 80,000 feet.

| | Initial | After life tests |
|---|---|---|
| Coil to case | N/A | 350 V rms |
| Aux contacts | N/A | |
| All other points | N/A | 350 V rms |

Dash numbers and general characteristics: See table I.

Maximum contact drop initial: 0.150 volt (dash numbers 1 and 3), 0.375 volt (dash numbers 2, 4, 5, and 6).

After life test: 0.175 volt (dash numbers 1 and 3), 0.375 volt (dash numbers 2, 4, 5, and 6).

Overload current: 20 amperes, 28 V dc.

Rupture current: 40 amperes, 115 V ac, 400 Hz.

Duty rating: Continuous.



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CONFORMANCE INSPECTION:

Performance of groups B and C tests are not applicable.

Group A acceptance reports shall be submitted to the qualifying activity on a yearly basis in order to retain qualification for this military specification sheet.

Qualification by similarity: See MIL-PRF-6106.

TABLE I. Dash numbers and characteristics.

| PIN number MS27245- | Type | Coil | Terminal type | Mounting socket | Auxiliary contacts | Max weight in pounds |
|---|---|---|---|---|---|---|
| 1 | I | dc | Plug-in | Flange | N/A | 0.08 |
| 2 | I | dc | Plug-in | Stud | N/A | 0.23 |
| 1/ 3 | I | dc | Plug-in | Flange | N/A | 0.08 |
| 4 | I | dc | Printed wire (PW) | Stud | N/A | 0.23 |
| 5 | I | dc | Solder hook | Stud | N/A | 0.23 |
| 6 | I | dc | Solder hook | Stud | N/A | 0.23 |

1/ All tests specified in each subgroup of the Group A inspection table of MIL-PRF-6106 are applicable to -3.

TABLE II. Operating characteristics.

| PIN MS 27245- | Coil data | | | | | | | | | | | Time - (milliseconds maximum) | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Coil | Nominal | | | Max | | Max pick-up voltage | | | Hold voltage 2/ | Drop out voltage 2/ | Operate 3/ | Release 4/ | Contact Bounce | | | |
| | | | | | | | | | | | | | | Main | | Aux | |
| | | Volts 1/ | Freq Hz | Ω Res ±10% | Volts | Amp | Normal 2/ | High temp test | Cont current test | | | | | NO | NC | NO | NC |
| MS27245-1 | X1,X2 | 28 | dc | 300 | 29 | .12 | 18 | 18 | 19.8 | 7.0 | 1.0 | 10 | 10 | 2 | 5 | --- | --- |
| MS27245-2 | X1,X2 | 28 | dc | 300 | 29 | .12 | 18 | 18 | 19.8 | 7.0 | 1.0 | 10 | 10 | 2 | 5 | --- | --- |
| MS27245-3 | X1,X2 | 28 | dc | 300 | 29 | .12 | 18 | 18 | 19.8 | 7.0 | 1.0 | 10 | 10 | 2 | 5 | --- | --- |
| MS27245-4 | X1,X2 | 28 | dc | 300 | 29 | .12 | 18 | 18 | 19.8 | 7.0 | 1.0 | 10 | 10 | 2 | 5 | 4 | 4 |
| MS27245-5 | X1,X2 | 28 | dc | 300 | 29 | .12 | 18 | 18 | 19.8 | 7.0 | 1.0 | 10 | 10 | 2 | 5 | --- | --- |
| MS27245-6 | X1,X2 | 28 | dc | 300 | 29 | .12 | 18 | 18 | 19.8 | 7.0 | 1.0 | 10 | 10 | 2 | 5 | --- | --- |

1/ Caution: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With rated coil voltage.
4/ From rated coil voltage.



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TABLE III. Rated contact load (amperes per pole) (case grounded) (-1 and -3).

| Type of load | Life operating cycles x $10^3$ | 28 V dc | | | | 115 V ac, 1 phase | | | | 115/200 V ac, 3 phase 1/ | | | | See appropriate notes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Main | | Aux | | Main | | Aux | | Main | | Aux | | |
| | | NO | NC | NO | NC | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | |
| Resistive | 100 | 10 | 10 | | | 10 | | | | 10 | | | | |
| Inductive | 100 | 4 | 4 | | | 4 | | | | 4 | | | | |
| Inductive | | | | | | | | | | | | | | |
| Motor | 100 | 3 | 3 | | | 3 | | | | 3 | | | | |
| Lamp | 100 | 2 | 2 | | | 2 | | | | 2 | | | | |
| Transfer load | | | | | | | | | | | | | | 2/ |
| Mechanical life reduced current | 400 | 2.5 | 2.5 | | | | | | | | | | | |
| Mixed loads | 50 | | | | | | | | | | | | | 3/ |

1/ 115/200 V ac for 60 Hz ratings, absence of value indicates relay is not rated for 3-phase applications.
2/ Transfer load indicates relay is suitable for transfer between unsynchronized ac power supplies at rating indicated.
3/ Minimum current ratings 0.1 ampere resistive, 0.3 ampere inductive, and 0.5 ampere resistive are applicable in accordance with acquisition specifications except monitored to a 3 ohm contact resistance.

TABLE IV. Rated contact load (amperes per pole) (case grounded) (-2, -4, -5, and -6).

| Type of load | Life operating cycles x $10^3$ | 28 V dc | | | | 115 V ac, 1 phase | | | | 115/200 V ac, 3 phase 1/ (Not applicable to -2 and -6) | | | | See appropriate notes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Main | | Aux | | Main | | Aux | | Main | | Aux | | |
| | | NO | NC | NO | NC | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | |
| Resistive | 50 | 10 | 10 | | | 10 | | | | 10 | | | | |
| Inductive | 50 | 6 | 6 | | | 6 | | | | 4 | | | | |
| Inductive | | | | | | | | | | | | | | |
| Motor | 50 | 4 | 4 | | | 4 | | | | 4 | | | | |
| Lamp | 50 | 2 | 2 | | | 2 | | | | 2 | | | | |
| Transfer load | | | | | | | | | | | | | | 2/ |
| Mechanical life reduced current | 400 | 2.5 | 2.5 | | | | | | | | | | | |
| Mixed loads | 50 | | | | | | | | | | | | | 3/ |

1/ 115/200 V ac for 60 Hz ratings, absence of value indicates relay is not rated for 3-phase applications.
2/ Transfer load indicates relay is suitable for transfer between unsynchronized ac power supplies at rating indicated.
3/ Minimum current ratings 0.1 ampere resistive, 0.3 ampere inductive, and 0.5 ampere resistive are applicable in accordance with acquisition specifications except monitored to a 3 ohm contact resistance.



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Referenced documents. In addition to MIL-PRF-6106, this document references the following:

MIL-PRF-23586 SAE-AS22759/5 SAE-AS22759/11 ASTM-D5948

Custodians:
NAVY - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-2012-008)

Review activities:
Navy - AS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil/.



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